Today's topic:

How do we evaluate the quality
of a binary classifier or a medical test ?

Precision, Recall, ROC, AUC,
F1, Sensitivity, Specificity, etc.

ROC Curve & Precision-Recall Curve

Receiver Operating Characteristic (**ROC**) curves for a binary classifier:

True Positive Rate (TPR) vs **False Positive Rate (FPR)**
at various threshold settings.

ROC curve was first developed by radar engineers during **World War II** for detecting enemy objects.

TPR = REC = **Recall or Sensitivity** = TP/P = TP / (TP + FN)
(planes classified as planes / all planes' events)

FPR = **False Positive Rate** = **fall-out** = FP / N = FP / (FP + TN)
(noise classified as planes / all noise events)

PRE = **Precision** = TP/(TP+FP)
(planes classified as planes / all events classified as planes)

TNR = SPC = **Specificity** = TN/N = TN/(FP+TN)
(True Negative Rate)

AUC = Area Under the Curve

Simply the area under the ROC curve (1 = very good, 0.5 = bad)

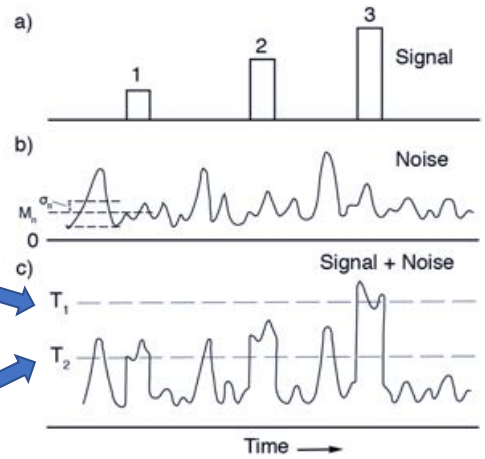
F1 Score - a single number evaluation metric, harmonic average of the **precision** and **recall** :

$$\frac{1}{\frac{1}{2}\left(\frac{1}{\text{recall}} + \frac{1}{\text{precision}}\right)}$$

Radar detects planes.
It need to separate
"plane" signals from noise.

Planes

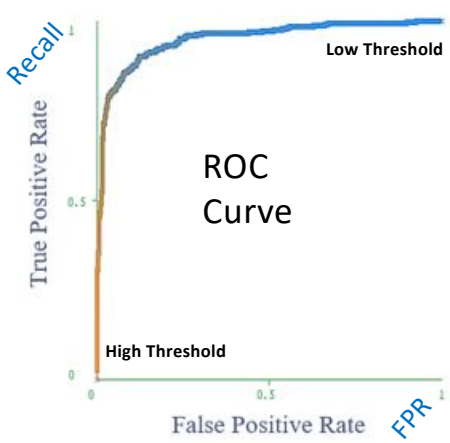
Noise

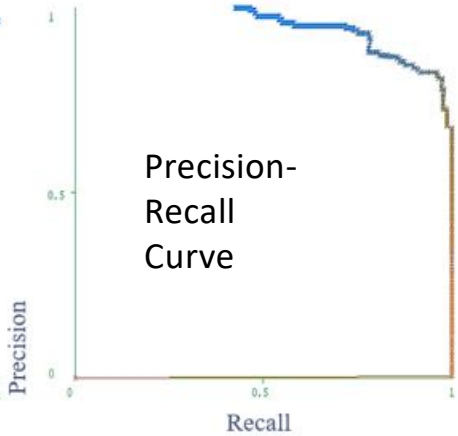


High Precision, Low Recall

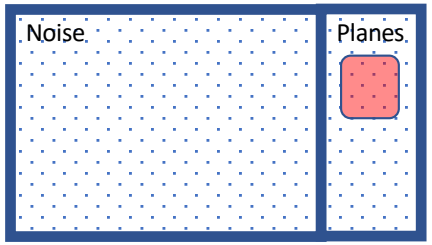
Selecting Threshold

Low Precision, High Recall



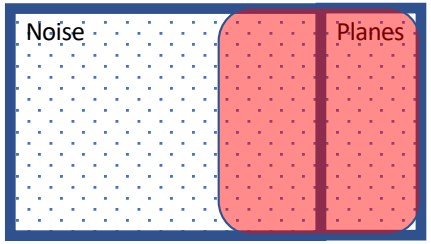


Noise **Planes**



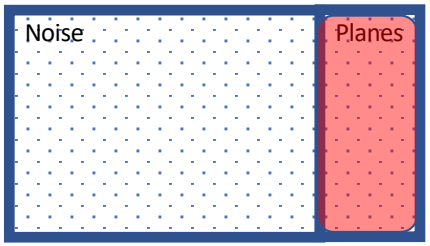
Precision – 100%
Recall – 15%

Good Precision = Sniper

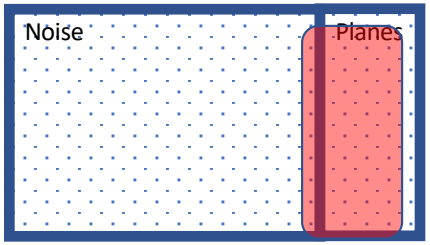


Precision – 50%
Recall – 100%

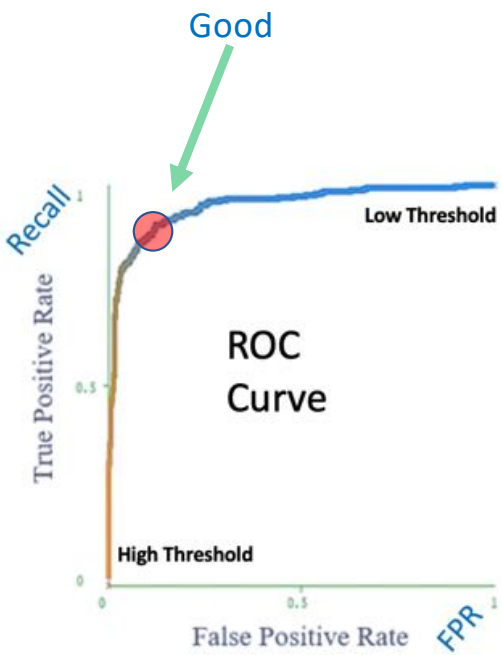
Good Recall = Cover All

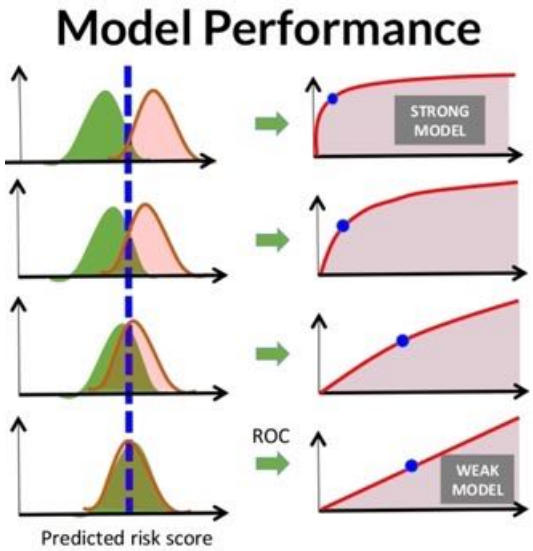


Ideal:
Precision – 100%
Recall – 100%



Good:
Precision – 90%
Recall – 90%





From:
https://www.slideshare.net/KristenHunter/civitas-learning-understanding-roc-curves

For measuring the performance of retrieval systems, several measures are used that depends on the following four major parameters:

```
True Positive  (TP): True  is identified True  (correct identification)
True Negative  (TN): False is identified False (correct identification)
False Positive (FP): True  is identified False (wrong identification)
False Negative (FN): False is identified True  (wrong identification)

P = TP + FN (number of correct identification cases)
N = FP + TN (number of wrong identification case) 25

Sensitivity = Recall = True Positive Rate (TPR) : TPR = (TP / P) = TP / (TP + FN)
Specificity (SPC) or True Negative Rate (TNR)   : SPC = (TN / N) = TN / (FP + TN)
Precision or Positive Predictive Value (PPV)    : PPV = TP / (TP + FP)
Negative Predictive Value (NPV)                 : NPV = TN / (TN + FN)
Fall-Out or False Positive Rate (FPR)           : FPR = (FP / N) = FP / (FP + TN) = 1 – SPC
False Discovery Rate (FDR)                      : FDR = FP / (FP + TP) = 1 – PPV
Miss Rate or False Negative Rate (FNR)          : FNR = FN / (FN + TP)
Accuracy (ACC)                                  : ACC = (TP+TN) / (P+N)
F1 Score                                        : F1  = 2*TP/(2*TP+FP+FN)

Mathews Correlation Coefficient (MCC):
    MCC=(TP*TN-FP*FN)/sqrt((TP+FP)*(TP+FN)*(TN+FP) *(TN+FN))

Informedness = Sensitivity + Specificity -1
Markedness   = Precision + NPV -1
```

From:
https://www.researchgate.net/post/How_do_I_calculate_the_false_alarm_rate_for_face_detection

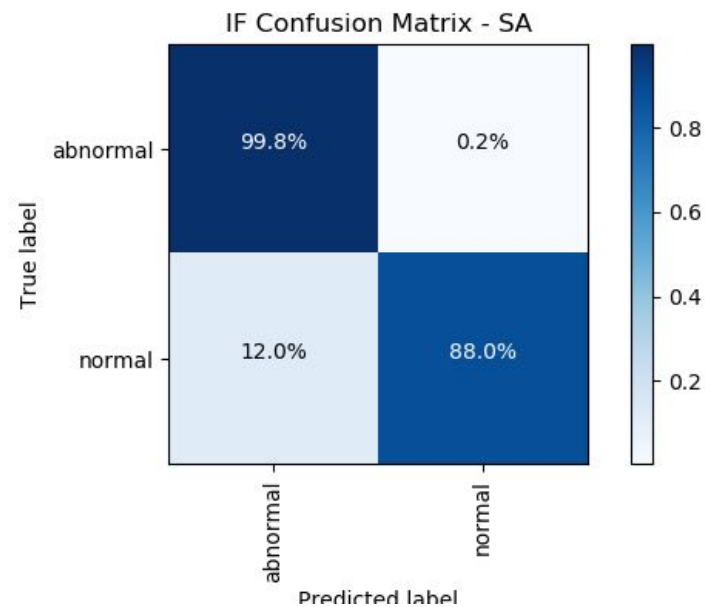
Confusion Matrix

For a simple binary classifier we are predicting one of two possibilities (Yes/No, Positive/Negative, Plane/No-Plane, Normal/Abnormal, etc.).

So we construct a 2x2 matrix to visualize True & False Positives and Negatives.

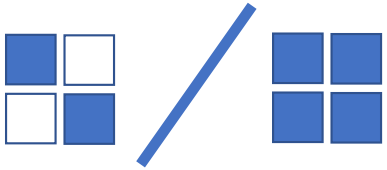
		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

```
from sklearn import metrics

expected = target
predicted = model.predict(data)

metrics.precision_score(expected, predicted)
metrics.recall_score(expected, predicted)
metrics.roc_curve(expected, predicted)
metrics.roc_auc_score(expected, predicted)

metrics.confusion_matrix(expected, predicted)

[[194156   4864]
 [    23    321]]
```



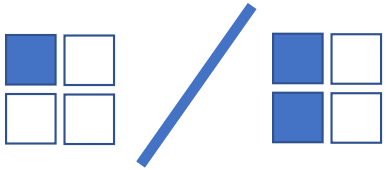
Accuracy = sum(Diagonal) / sum(ALL) = sum(Correct) / sum(ALL)
Precision = TP / (TP+FP)
Recall = TP / (TP+FN) = TP / All_Actual_Positives

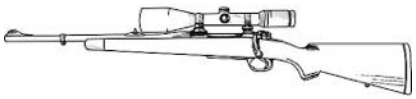
Accuracy = sum(Diagonal) / sum(ALL)



		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

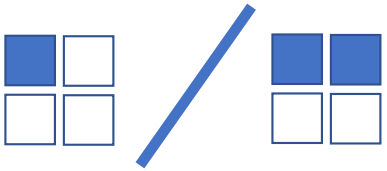
Precision = TP / (TP+FP) = TP / Predicted_Positives





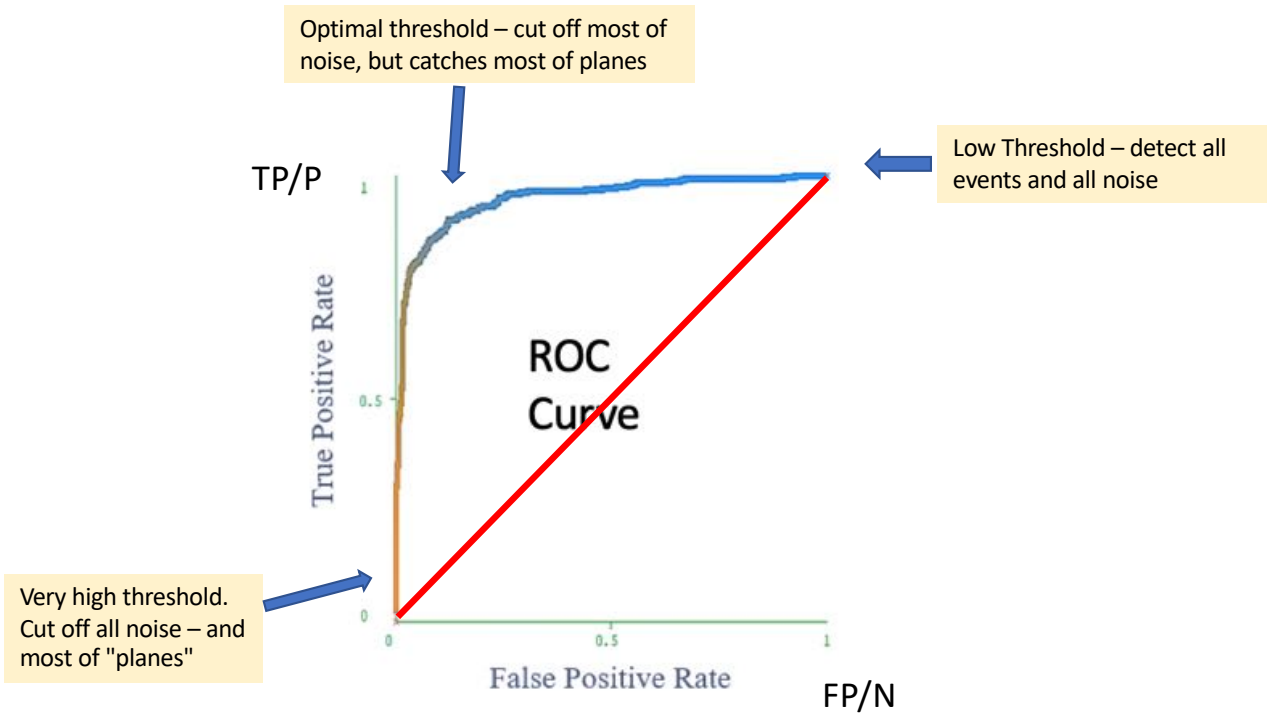
Sniper
- high precision (zero FP)
- low recall (can't get everything)

Recall = TP / (TP+FN) = TP / Actual_Positives



Nuclear Bomb
- low precision (many FP)
- high recall (get everything)

Good ROC curve is above diagonal



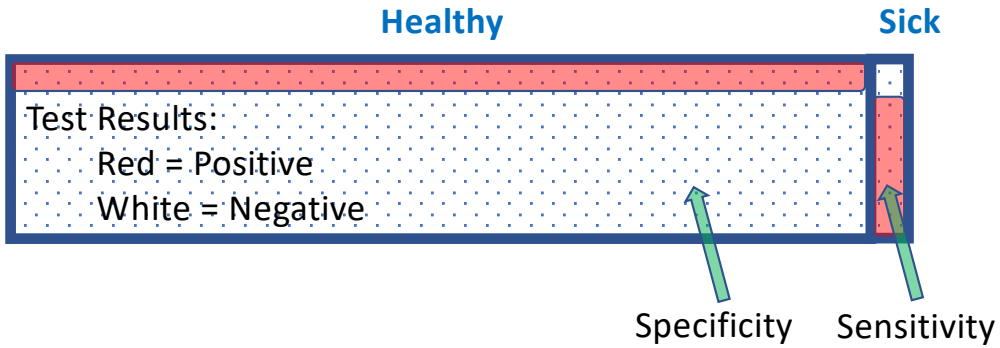
ROC curve axes:

vertical = True Positives / All Actual Positives

horizontal = False Positives / All Actual Negatives

Medical Test Paradox – Predictive Power of Positive or Negative Test Result

from https://www.youtube.com/watch?v=lG4VkPoG3ko



We have 1,000 people.
1% of them have a disease:

```
N_sick    = 10
N_healthy = 990
```

We use a test with **sensitivity SE** = 90% and **specificity SP** = 91%.

This means that test shows results as following:

```
TP = True Positive   =   N_sick    * SE/100   =     9 out of  10
TN = True Negative   =   N_healthy * SP/100   =   901 out of 990

FN = False Negative  =   1 out of  10
FP = False Positive  =  89 out of 990
```

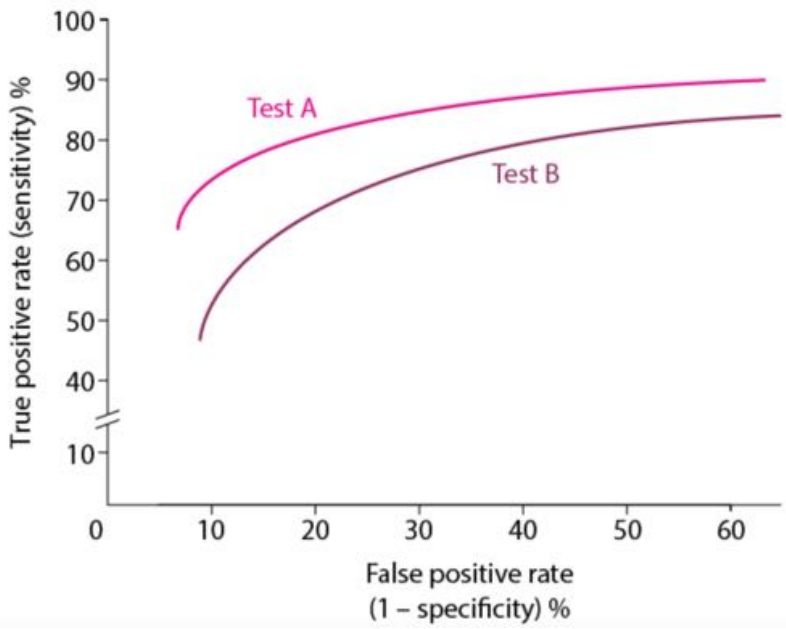
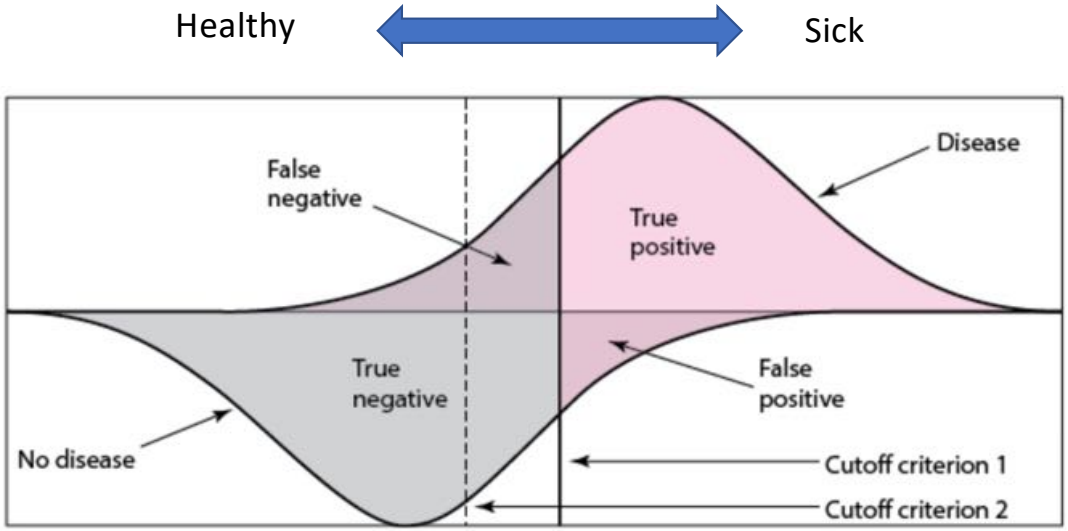
	Positive	Negative
Disease	9	1
Healthy	89	901

What is the predictive power of this test?

```
Positive result :  TP/(TP+FP)  =    9/(9+89)   =  0.092  (1 in 11 chance)

Negative result :  TN/(TN+FN)  =  901/(901+1)  =  0.999  (very good)
```

```
Sensitivity = Recall = True Positive Rate (TPR) : TPR = (TP / P) = TP / (TP + FN)
Specificity (SPC)    = True Negative Rate (TNR) : SPC = (TN / N) = TN / (FP + TN)
```

Medical Tests

Merck Manuals:
https://www.merckmanuals.com/professional/special-subjects/clinical-decision-making/understanding-medical-tests-and-test-results

Typical ROC Curves